

Reconditioning of Cylinder Block – Cleaning of Metal Spray Waste

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

This Best Practice is a result of Six Sigma methodology at Ferreyros' Machine Shop. After the final machining of top deck, we need to clean the metal spray waste from water passage holes, threaded holes, dowel holes, and corners of the top deck.

Previously, reconditioned block with metal spray was cleaned using different file types and manual threaded taps. The job required 16 man-hours. Currently, using different kinds of tools the effective time is 9 man-hours.

Both former and current procedures of cleaning will be described below.

2.0 Best Practice Description

Previously, the cylinder block cleaning process after final machining was using flat file, circle file and manual threaded tap. The process shown below:

N°	Step	Picture	Operators required for both top deck	Time for both top deck
1	Cleaning inspection camshaft window		2	45 min
2	Cleaning curve area		2	16 min
3	Cleaning surrounding surfaces		2	30 min

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4	Cleaning dowel hole		2	60 min
5	Cleaning water passages and threaded holes		2	160 min
6	Side and bottom threaded holes		2	83.5 min

Currently, for the cleaning of metal spray waste, the following is required:

- Special taps
- Special drill cutting tool
- Cylindrical and conical cutting tool
- Mounted stones

The technique is simple: install the cylindrical or conical cutting pneumatic emery tool to perform cleaning camshaft inspection windows, curved areas and adjacent surfaces. Then, for cleaning dowel holes, mounted stones are used. Next step is to clean cylinder head holes for which a device is used as a guide and special drill cutting tool. Finally in cleaning side and bottom holes a threaded is used with special tap (machine tap).

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Works involved are:

- Cleaning dowel holes & threaded holes: 80 min per cylinder block
- Removing burrs: 30 min per cylinder block
- Bottom cylinder block: 30 min
- Sanding (finishing): 30 min

3.0 Implementation Steps

- Define the areas to be cleaned.
- Use the toolkit as detailed in subsection 5 “Resources Required”
- Cleaning Process:
 - Cleaning camshaft windows
 - Cleaning curved areas
 - Cleaning dowels areas
 - Cleaning adjacent surfaces
 - Cleaning cylinder head holes area
 - Cleaning threaded holes

4.0 Benefits

The main benefit of this Best Practice is to reduce cylinder block cleaning time, the following graphs show the cleaning time before and after.

- Reduction of man-hours by 44%
- Less time for cylinder blocks cleaning. Old process 16 hours, current process 9 hours.
- Work improvement, more effective working with technology that facilitates the tasks.

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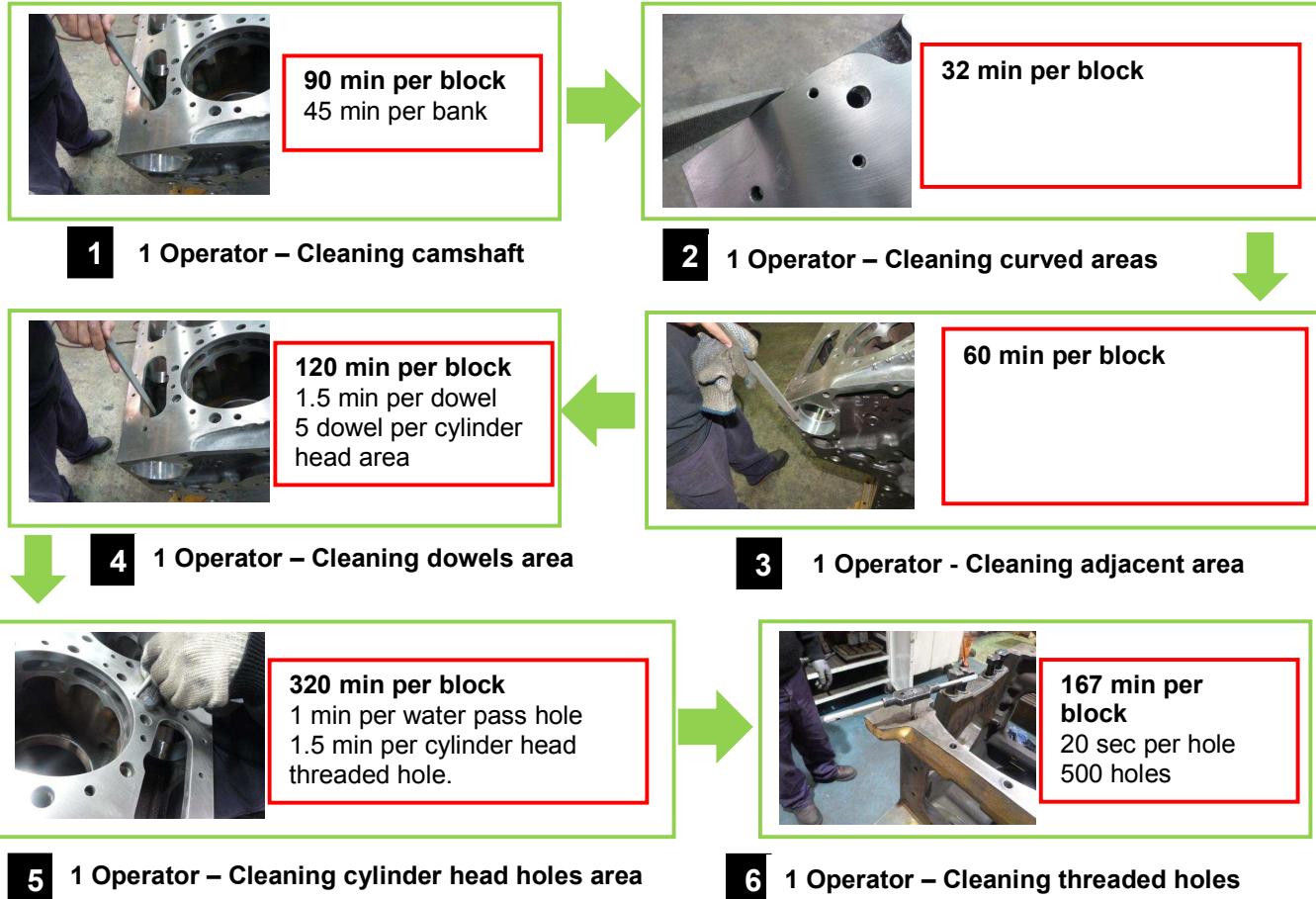
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Old Process Method



Total processing time: 16 hours

* Total processing time include time needed

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Current Process Method



60 min per block
30 min per bank

1**1 Operator – Cleaning camshaft**

20 min per block

2**1 Operator – Cleaning curved areas**

80 min per block
1 min per dowel
5 dowel per
cylinder head area

4**1 Operator – Cleaning dowels area**

40 min per block

3**1 Operator - Cleaning adjacent area**

160 min per block
10 min per cylinder head
threaded hole.

5**1 operator – Cleaning cylinder head holes area**

42 min per block
5 sec per hole
500 holes

6**1 Operator – Cleaning threaded holes**

Total processing time: 9 hours

* Total processing time include time needed

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5.0 Resources Required

Quantity	Tools
1	Stanley 019151M 19-inch Series 2000 Tool Box with Tray
1	Multi-Tool system: Dremel 8200
2	Special threading tap OSG 5/16-18 COATED - Harmetall
2	Special threading tap OSG 3/8-16 COATED - Harmetall
2	Special threading tap OSG 5/8-11 COATED - Harmetall
2	Special threading tap 3/4-14 NPT COATED - Harmetall
2	Special threading tap OSG 1/2-13 COATED - Harmetall
2	Special threading tap OSG 3/4-16 COATED - Harmetall
2	Universal Tool UT8729-18 ANGEL DIE DRINDER, 1800RPM, REAR EXHAUST, 0.9 HP
1	Drill INGERSOL RAND 8506 ATC
2	Special drill cutting tool 20.6 Techcut Carbide - Harmetall
2	Special drill cutting tool 11.5 Techcut Carbide - Harmetall
2	Cylindrical cutting tool ZYA 1225 Carburo de Tugsteno 12 mm
2	Cutting tool KEL 1020 Carburo de Tugsteno
1	Hand Deburring Tool 155-00184

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Name Roger Rodriguez Ocrospoma

E mail roger.rodriguez@ferreyros.com.pe

Recognition:

Team: Miguel Mauriola / Operations Support Chief
 Carlos Espinoza / Technical Communicator
 José Huayamis / Technical Communicator
 Sandro Falcón / CNC programmer
 Roger Rodriguez / Operations Support Assistance

Owner: Rafael Volta / Machine Shop Chief

Sponsor: Luis Aguilar / Manager Machine Shop

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